

Kinora Lumbar Spine Clinical Database

Clinical AI Orientation (Part 23)

L1–L5, Discs, Ligaments, Canister Muscles, Plexuses, Pathologies incl. Cauda Equina
(complements Part 5)

Version 1.0 — Kinora Admin Conocimientos Upload

Disclaimer

*Educational. Complements Part 5 lumbar orientation. **CAUDA EQUINA and progressive neuro deficit are emergencies.***

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1. Vertebrae

RECORD: L1

ID: LUM-V-1

Vertebra: L1

Landmarks: Large body, broad SP, mamillary processes, TP

Facet_Orientation: Progressively more sagittal caudal; L5–S1 transitional

Clinical: L4–5 / L5–S1 highest pathology burden

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: L2

ID: LUM-V-2

Vertebra: L2

Landmarks: Large body, broad SP, mamillary processes, TP

Facet_Orientation: Progressively more sagittal caudal; L5–S1 transitional

Clinical: L4–5 / L5–S1 highest pathology burden

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: L3

ID: LUM-V-3

Vertebra: L3

Landmarks: Large body, broad SP, mamillary processes, TP

Facet_Orientation: Progressively more sagittal caudal; L5–S1 transitional

Clinical: L4–5 / L5–S1 highest pathology burden

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: L4

ID: LUM-V-4

Vertebra: L4

Landmarks: Large body, broad SP, mamillary processes, TP

Facet_Orientation: Progressively more sagittal caudal; L5–S1 transitional

Clinical: L4–5 / L5–S1 highest pathology burden

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: L5

ID: LUM-V-5

Vertebra: L5

Landmarks: Large body, broad SP, mamillary processes, TP

Facet_Orientation: Progressively more sagittal caudal; L5–S1 transitional

Clinical: L4–5 / L5–S1 highest pathology burden

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

2. Discs

RECORD: Lumbar Discs L1–2 to L5–S1

ID: LUM-DISC-01

Nomenclature: Protrusion/extrusion/sequestration; Pfirrmann degeneration grades; Modic endplate

Herniation: Posterolateral typical; may compress traversing root at L4–5 (L5 root) etc.

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

3. Ligaments

RECORD: ALL/PLL/Flavum/Interspinous/Supraspinous/Iliolumbar

ID: LUM-LIG-01

Iliolumbar: Stabilizes L5 on ilium; resists anterior shear

Flavum_Hypertrophy: Stenosis contributor

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

4. Muscles & Canister

RECORD: Multifidus

ID: LUM-MUS-01

Role: Segmental stability; atrophy common in LBP

Rehab: Local activation then functional

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Erector Spinae

ID: LUM-MUS-02

Role: Global extension/moment resistance

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Quadratus Lumborum

ID: LUM-MUS-03

Role: Frontal plane; hip hike; breathing accessory

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Psoas Major

ID: LUM-MUS-04

Role: Hip flexor; lumbar stabilizer debate; crosses lumbar

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Abdominals (RA/EO/IO/TA)

ID: LUM-MUS-05

Role: IAP canister with diaphragm/PF

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Diaphragm & Pelvic Floor Integration

ID: LUM-MUS-06

Role: Canister model; coordinate breath + continence + load

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

5. Nerves

RECORD: Lumbar & Sacral Plexus / Neuro Screen

ID: LUM-NRV-01

Dermatomes: L2 ant thigh; L3 medial knee; L4 medial leg; L5 dorsal foot/1st web; S1 lateral foot

Myotomes: L2–3 hip flex; L3–4 knee ext; L4 DF; L5 great toe ext; S1 PF

Reflexes: Patellar L3–4; Achilles S1–2

Cauda_Equina: EMERGENCY — retention, saddle anesthesia, bilateral deficits

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

6. Pathologies

RECORD: Disc Herniation / Sciatica

ID: LUM-PATH-01

Care: Most improve; guide persistent/progressive; surgery selected

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Stenosis

ID: LUM-PATH-02

Clues: Neurogenic claudication relief with flexion

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Facet Syndrome

ID: LUM-PATH-03

Clues: Extension-rotation pain

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Spondylolysis / Listhesis

ID: LUM-PATH-04

Clues: Young athlete extension pain; step-off

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: DDD

ID: LUM-PATH-05

Note: Age-related; correlate clinically not MRI alone

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: SI Referral

ID: LUM-PATH-06

Note: Laslett cluster; see Pelvis Part 6

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Cauda Equina

ID: LUM-PATH-07

Action: Emergency MRI/surgical decompression pathway

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Compression Fracture

ID: LUM-PATH-08

Clues: Osteoporosis, trauma, sitting intolerance

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.